Australian National University

LLMs Unplugged | Cybernetic Studio

Weighted Randomness

Learn how to make random choices where some options are more likely than others — an operation at the core of all generative AI.

You will need

- 10-sided dice (d10)
- coloured marbles or beads in a bag

Your goal

To randomly choose from a fixed set of outcomes according to a given probability distribution.

Key idea

Sometimes we need to make random choices where some outcomes are more likely than others. There are ways to do this which ensure certain relationships *on average* between the outcomes (e.g. one outcome happening twice as often as another one).

Algorithm 1: beads in a bag

- **materials**: coloured beads, bag
- **setup**: count out a number of beads corresponding to the desired weights for each outcome
- **sampling procedure**: shake the bag, then draw one bead without looking

Example

You want to choose an ice cream flavour: `vanilla` 50% of the time, `chocolate` 30%, and `strawberry` 20%.

- add 5 white beads to the bag (corresponding to `vanilla`)
- add 3 brown beads to the bag (corresponding to `chocolate`)
- add 2 red beads to the bag (corresponding to `strawberry`)

Draw a bead from the bag — that's your ice-cream choice for today.

Algorithm 2: dice with ranges

- **materials**: d10 (or d6, d20 as alternatives)
- **setup**: assign number ranges proportional to weights (see table, right)
- **sampling procedure**: roll the die, then look up the corresponding outcome

Example

- for 60% vanilla/40% chocolate, roll a d10: 1-6 means `vanilla`, 7-10 means `chocolate`
- for 50% vanilla/30% chocolate/20% strawberry, roll a d10: 1-5 means `vanilla`, 6-8 means `chocolate`, 9-10 means `strawberry`

You can use different dice (d6, d10, d20, d120, etc.), it will just change the number ranges corresponding to each outcome.

d10 dice roll → outcome mapping table

	1	2	3	4	5	6	7	8	9	10
2	0–4	5–9								
3	0–3	4–6	7–9							
4	0–2	3–5	6–7	8–9						
5	0–1	2–3	4–5	6–7	8–9					
6	0	1	2	3	4	5	6	7	8	9
7	0	1	2	3	4	5	6	7	8	9
8	0	1	2	3	4	5	6	7	8	9
9	0	1	2	3	4	5	6	7	8	9